

Software Engineering - Lab 3

Component Modelling & Architectural Pattern Selection

Book Club Reading Challenge & Discussion Portal

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Architecture Selection

We chose Layered Architecture for the Book Club Reading Challenge & Discussion Portal.

Reason 1 - Separation of concerns: The presentation, business, and data responsibilities can be separated into clear layers. This fits the portal because member interaction, reading-challenge/discussion logic, and stored portal data have distinct responsibilities.

Reason 2 - Maintainable component structure: The existing requirements cover reading challenges, progress tracking, discussion creation/participation/moderation, and authenticated member functions. A layered structure keeps these related business functions separate from the user interface and database, making the component interactions clearer and easier to maintain.

Security Advantage: The Authentication Component provides a dedicated access-control boundary. This supports the requirement that authenticated users must log in before accessing member-specific reading and discussion functions.

Performance Benefit: Separating presentation, business logic, and data access allows each layer to handle its own work without mixing responsibilities. This supports the portal's response-time requirement for reading-challenge and discussion pages and helps keep the interface responsive under normal operating conditions.

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Component Diagram - Layered Architecture

